Do Now · retrieval

Lesson 116 · Linkers, Loaders and Program Libraries

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which compilation stage converts source characters into tokens?

Q2

In a BST, where is a value less than the current node placed?

Q3

Why can an array provide direct indexed access?

Q4

A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Compile modules — Source files become object modules with symbols/references.

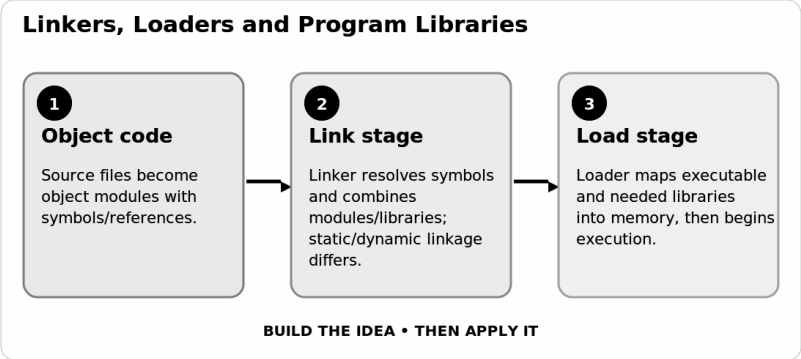
Core idea 2

Link dependencies — Linker resolves symbols and combines modules/libraries; static/dynamic linkage differs.

Core idea 3

Load and run — Loader maps executable and needed libraries into memory, then begins execution.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

linker — Resolves symbols and combines object modules.

loader — Places an executable in memory ready to run.

library — Reusable compiled routines/functions.

dynamic linking — Resolution/use of shared code at load time or runtime.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

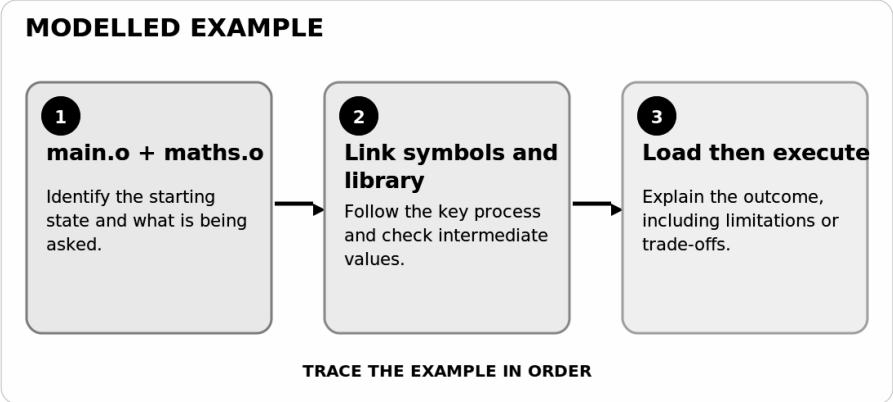
Problem

main.c calls a function in maths.c. Explain the steps from separate source files to a running program.

Follow the worked method

1. Model answer — Compile source modules into object files; linker resolves the call and produces linked executable (or dynamic references); loader places executable and required libraries in memory and starts execution. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain the roles of linker and loader with examples. [6]

Question 2

Distinguish static from dynamic linking, giving one trade-off of each. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A deployed app fails because a required shared library is unavailable. Explain why compile success did not prevent this. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A successful compilation guarantees that every dynamically linked library will be available at runtime.

Use key vocabulary from memory

1. Explain linker in your own words.

2. Explain loader in your own words.

3. Explain library in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.